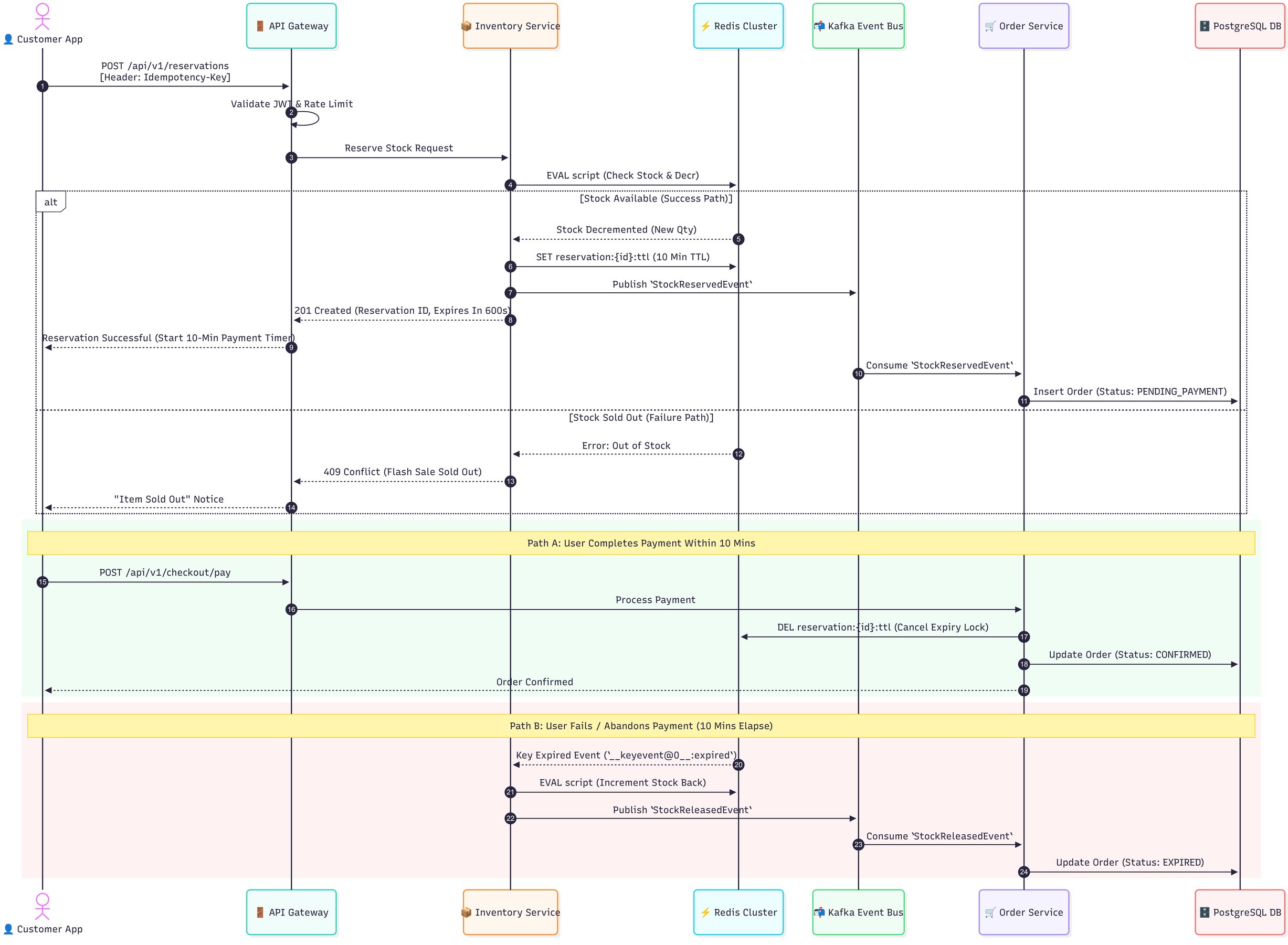

👤 Customer App
🚪 API Gateway
📦 Inventory Service
⚡ Redis Cluster
📬 Kafka Event Bus
🛒 Order Service
🗄️ PostgreSQL DB
POST /api/v1/reservations [Header: Idempotency-Key]
Validate JWT & Rate Limit
Reserve Stock Request
EVAL script (Check Stock & Decr)
alt
[Stock Available (Success Path)]
Stock Decremented (New Qty)
SET reservation:{id}:ttl (10 Min TTL)
Publish 'StockReservedEvent'
201 Created (Reservation ID, Expires In 600s)
Reservation Successful (Start 10-Min Payment Timer)
Consume 'StockReservedEvent'
Insert Order (Status: PENDING_PAYMENT)
[Stock Sold Out (Failure Path)]
Error: Out of Stock
409 Conflict (Flash Sale Sold Out)
"Item Sold Out" Notice
Path A: User Completes Payment Within 10 Mins
POST /api/v1/checkout/pay
Process Payment
DEL reservation:{id}:ttl (Cancel Expiry Lock)
Update Order (Status: CONFIRMED)
Order Confirmed
Path B: User Fails / Abandons Payment (10 Mins Elapse)
Key Expired Event ('__keyevent@0__:expired')
EVAL script (Increment Stock Back)
Publish 'StockReleasedEvent'
Consume 'StockReleasedEvent'
Update Order (Status: EXPIRED)